UNIVERSITY OF UYO

FACULTY OF SCIENCE

DEPARTMENT OF MATHEMATICS

FIRST SEMESTER EXAMINATION SESSION 2022/2023

COURSE CODE: MTH211 (MATHEMATICAL METHODS 1) TIME: 2HOURS

INSTRUCTION: ANSWER ALL QUESTIONS

1. Prove that $\lim_{(x,y)\to(1,2)} (x^2y - 3x + 2y - 2) = 1$.

2. Find the value of the point $a \in [3,5]$ at which the continuously differentiable function $f(t) = 2t^2 - 7t + 10$ satisfies the Mean-value theorem on $[3,5]$.

3. If $Z = Z(U,V)$, $U = r^2 - s^2$ *and* $V = 2sr$. Show that

$$r\frac{\partial z}{\partial s} + s\frac{\partial z}{\partial r} = 2(r^2 + s^2)\frac{\partial z}{\partial v}$$

4. If $u^2 - v = 3x + y$ and $u - v^2 = x - 2y$, find u_x and v_y

5. Obtain the approximate value of $\sqrt{35}$ using differential

6. Obtain the Taylor's series polynomial of the function $f(x,y) = \cos(xy)$ about $(\frac{\pi}{3}, 1)$ up to the quadratic term

7. Find the directional derivative of the function $f(x,y,z) = 2x^2y + 3xyz + yz^2$ at point $(2,1,-3)$ in the direction of $2i - 3j + 4k$

8. Find the extrema of the function $f(x,y) = xy$ subject to $g(x,y) = x + y - 1$

9. Evaluate the curl of the vector $xyzi + 3xyj + (xz^2 - y^2z)$ at the point $(2, -1, 1)$

10. Find the work done by a force $F = 3i + 4j + 5k$ in moving a particle from point P(2, 1, 1) to point Q(3, 6, 4), where all parameter are given in SI unit